

Professional Portfolio

Kalyani Guggilla | Product Management

Professional Bio

Kalyani Guggilla is a product manager focused on AI-powered, customer-facing products. With a background in software engineering and almost 6 years of experience in Boston, I specialized in translating emerging technologies such as natural language processing and generative AI into products that balance innovation with operational discipline. Currently completing graduate coursework in AI and machine learning leadership (AIML-505), I'm building the strategic, ethical, and technical fluency required to lead cross-functional teams in shipping trustworthy AI products.

Value Proposition and Audience

- **Personal Value Proposition:** I bridge the gap between technical AI capability and business outcomes turning emerging technologies into product strategies that are grounded in evidence, differentiated in the market, and governed responsibly. This artifact demonstrates that value by showing how I evaluate an emerging AI trend, connect it to measurable business results, and plan for both its upside and its risks before committing a roadmap to it.
- **Target Audience:** Hiring managers, product leaders, and directors of product in organizations building AI-driven or data-intensive products. This portfolio is relevant to them because it demonstrates the exact judgment a product manager must exercise when deciding whether and how to invest in an emerging AI capability: separating hype from evidence, defining a narrow and measurable use case, and building the guardrails that make an AI product trustworthy at scale.

Artifact 1

Artifact Title

Emerging AI Trends in Practice: Natural Language Processing and the Future of Customer Engagement

Introduction

This artifact is an individually authored research and future-casting report produced for the AIML-505 course. It examines natural language processing (NLP) specifically the shift away from simply producing fluent text and toward systems that reason over trusted data, hold context, personalize their replies, and act on a user's behalf and applies that trend to a realistic retail scenario. Using a fictional mid-market omnichannel retailer, "Verano Living," and a ChatGPT-based assistant named "Vera," the report translates an abstract technology trend into a concrete product concept, supported by peer-reviewed evidence and paired best-case and worst-case adoption scenarios.

Artifact Description

- **Objective:** To evaluate an emerging AI trend, ground it in credible evidence, and assess its practical product implications including its risks so that a leadership audience can make an informed adoption decision.
- **Process:** I selected NLP from a set of surveyed trends, then narrowed the focus to a specific product (ChatGPT), a specific application (retail customer engagement), and a specific analytical lens (best- and worst-case outcomes). I gathered and vetted sources, deliberately separating peer-reviewed findings from vendor marketing, and synthesized the research into a concrete product concept and two divergent future-casting narratives. I closed with a reflection on how prompt refinement improved the work and what disciplined AI adoption requires. This was an individually created artifact.
- **Tools and Technologies Used:** Generative AI (ChatGPT) as a research aid; scholarly databases and peer-reviewed literature; retrieval-augmented generation (RAG) and agentic-AI concepts as the subject matter; and APA 7th-edition formatting.

Artifact-Specific Value Proposition

- **Unique Value:** This artifact showcases product-management judgment rather than technical implementation alone: the ability to identify where value is created in an emerging technology (grounding, action, and personalization rather than fluency, which has become table stakes), to defend that view with evidence, and to plan for risk through scenario analysis. It demonstrates strategic foresight, evidence-based prioritization, and ethical and governance awareness core competencies for a product manager working with AI.
- **Relevance:** It aligns directly with my value proposition of bridging technical capability and business outcomes. Product managers are repeatedly asked to decide whether to invest in a new AI capability; this artifact is a worked example of exactly that decision, made with the discipline a hiring manager would want to see in a product leader.

Customization for Audience

- **Adaptations Made:** For a product-management audience, I framed the technology in terms of customer outcomes and business metrics conversion, average order value, return rates, and support cost rather than model internals. I anchored the analysis in a concrete product concept (the "Vera" assistant) with a defined feature scope, and I made the risk section explicit and actionable (grounding, governance, privacy safeguards, and human oversight) so it reads as a product-readiness checklist rather than an academic caveat.
- **Relevance:** These adaptations make the artifact more pertinent to product leaders because they mirror how product decisions are actually justified by tying capability to measurable outcomes and by showing that risks have been anticipated and mitigated before scaling.

Reflection

- **Significance:** I chose this artifact because it captures the full arc of product thinking around an emerging technology: understanding the technology, connecting it to business outcomes, and building the guardrails that make it trustworthy. It demonstrates that I can operate at all three levels at once rather than stopping at technical novelty the exact judgment a product manager is hired to provide.
- **Lessons Learned:** The most important lesson was that precision drives quality: refining a broad question into a specific actor, application, and lens produced far sharper analysis, and the same discipline separates organizations that benefit from AI from those that merely chase it. I also deepened my understanding of retrieval-augmented generation and AI governance, and of how to distinguish genuine peer-reviewed evidence from confident-sounding vendor marketing a skill directly applicable to vetting vendor claims on a real product roadmap.

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